

Gavin Xue

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TECHNICAL SKILLS

Languages: Python, JavaScript/TypeScript, C/C++, Java, SQL

Technologies: React, Node.js, Express.js, Next.js, Flask, MongoDB, PostgreSQL, Pandas, Matplotlib

Developer Tools: Git, Linux, Jira, Postman, Selenium, Cypress

EDUCATION

University of Waterloo

GPA: 89%

Candidate for the Bachelor in Computer Science

Sep 2023 – Apr 2028 (Expected)

- Awarded **1 of 10** René Descartes National Scholarships (**\$20,000**) from the Faculty of Mathematics
- Relevant courses: Algorithms Design and Data Abstraction, Object-Oriented Software Development

EXPERIENCE

Software Developer

Jan 2025 – Present

Environment and Climate Change Canada

- Led the development of a new climate model visualization product, utilizing large image datasets generated with Cartopy and **Matplotlib** to display for researchers and the public
- Refactored legacy code to align with object-oriented design principles, reducing the codebase by **500+** lines

Software Developer

Oct 2024 – Present

Develop for Good

- Built a responsive map application for Forgotten Felines of Sonoma County, using **Next.js** and Leaflet to improve the visualization of client data points and simplify administrative management
- Reduced data preprocessing times by **90%** by leveraging a fine-tuned OpenAI GPT-4 mini model for batch cleaning of **5,000+** entries, streamlining data preparation processes and improving efficiency
- Designed micro front-end architecture using Tailwind CSS and shadcn/ui components

Web Developer

May 2024 – Aug 2024

UW Mathematics Society

Waterloo, ON

- Refactored an Express.js server to implement dynamic route handling for **150+** store items, enhancing maintainability through a custom content management system
- Structured JSON data with schemas and **TypeScript** to ensure code base consistency

Information Technology Support Assistant

May 2024 – Aug 2024

University of Waterloo

Waterloo, ON

- Reduced testing times by automating manual processes with **Selenium Web Driver**, resulting in **up to 10x** faster execution speeds and significantly reducing labour hours
- Automated spam email detection using **Python scripts** to send requests to the Jira REST API, reducing the time required to clean the ticketing system by **80%** and boosting team productivity
- Led the development of a research group website, communicating with clients as needed to meet their requirements

PROJECTS

Item Lending System 📄 | React, Express.js, Node.js, PostgreSQL, Cypress

- Implemented **RESTful** APIs using Express.js, integrating the app with a cloud-based **PostgreSQL** database to facilitate CRUD operations for **300+** data entries
- Configured and secured email authentication using SMTP and **OAuth2** with UWaterloo's email server
- Minimized **end-to-end** test execution times, using **Cypress** for **over 90%** coverage across the site

Broken Link Checker 📄 | Python, Flask, BeautifulSoup4, Heroku

- Extracted HTML elements from WCMS 3 sites using BeautifulSoup4 for web-scraping, analyzing URL validity through HTTP response status codes
- Architected a bidirectional **websocket** connection for real-time updates between the **Flask** backend and the client side